Industrial–Logistics Coupling in Otay Mesa

A Spatial Analysis of Industrially Relevant Land, Ports of Entry, and Freight Corridors in San Diego

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SAN DIEGO
OTAY MESA
Brown Field
Municipal Airport
EAST OTAY MESA
OTAY MESA
Proposed Port of Entry, Otay II
United States
Mexico
United States
Mexico
Aeropuerto
Internacional de Tijuana
Mexico

Background

- Otay Mesa is a major industrial and logistics hub along the US-Mexico international border.
- The Otay Mesa Port of Entry supports regional and international trade.
- Industrial businesses often locate near freight infrastructure to reduce transportation costs.
- This study compares industrial businesses in Otay Mesa to those elsewhere in San Diego County.
- **Are industrial businesses in Otay Mesa more closely associated with freight infrastructure and ports of entry than industrial businesses elsewhere in San Diego County?**

Literature Review

Why does Otay Mesa look like a border-industrial node?

Content:

- Transport infrastructure shapes industrial location.
- Firms and logistics activities tend to cluster near transport nodes, corridors, and market-access points.
- Border regions intensify this logic. POEs and cross-border trade flows make accessibility especially important for manufacturing, wholesale, and logistics activities.
- Local land-use regulation channels the pattern. Zoning, General Plan designations, and IBT/border industrial policy define where these activities can legally and administratively concentrate.
- The puzzle: **Is industrial activity spatially organized around border-serving freight infrastructure?**

Inspiration:

This motivates our two-track design: test spatial association with GIS, then interpret the pattern through planning and regulatory context.

Data Used

Three Layers of Evidence

The main data design combines City of San Diego land-use layers, Census tract-level industrial employment proxies, and SANDAG FreightViewer infrastructure layers. Census tracts or block groups provide the common spatial unit for the baseline analysis. NAICS-filtered business points are used as an establishment-level validation layer, while GADM or other boundary datasets are used only as reference geography when regional context is needed.

Methodology

From policy geography to measurable spatial layers

Planning rules

- Zoning
- General Plan Land Use
- Community Planning Districts

Industrial activity

- Census tract industrial employment proxy

- Optional NAICS-filtered business points

Freight access

- SANDAG FreightViewer POEs
- Major roads
- Truck inspection / parking
- Freight rail

Key design choice:
Tracts provide a stable public baseline; business points provide finer validation.

GIS provides measurable indicators; planning and infrastructure evidence defines the plausible causal pathway.

Main takeaway:
Our method separates measurement from interpretation: GIS tests spatial association, while planning and infrastructure history support the mechanism.

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Findings

Map of Industrial Businesses in San Diego County

Finding 1 - Identifying Industrial Businesses

The largest category, wholesale trade (42), comprises ~39% of all industrial businesses in San Diego County.

Finding 2 - Distance Measurements Between Study Areas & Major Roads

Industrial businesses in Otay Mesa are approximately 62% closer to major freight roads than businesses elsewhere in San Diego County.

On average, industrial businesses in Kearny Mesa and Otay Mesa have the shortest distances to a major transportation corridor.

Finding 3 - Distance Measurements Between Study Areas & Ports of Entry

Distance to nearest Port of Entry

- Otay Mesa businesses are a median of 1.09 miles from a port, compared to 21–25 miles for every other study area.
- Every single business in every subarea has Otay Mesa POE as its nearest crossing.

Finding 4 - Overlap Between Study Areas & Freight Corridor Buffers

- Otay Mesa and Kearny Mesa have strong freight road proximity at ½ mile (81.7% and 90.3%), and near-total coverage at 1 mile (98.6% and 100%).
- Miramar sits in the middle, with about half its businesses (48.1%) reaching a freight road within ½ mile, but a third remain beyond 1 mile.
- Sorrento Valley is the clear outlier, as only 3.3% within ¼ mile, and just 17.6% even within 1 mile, meaning over 80% of its industrial businesses operate far from any major freight corridor.

Finding 5 - Overlap Between Industrial Land Uses, Employment Centers, and Freight Infrastructure

Main Findings Summary

We identified **2,868 Industrial Businesses in San Diego.**

We found that **Otay Mesa's** industrial businesses are **~62% closer to a major road** than similar businesses elsewhere in San Diego.

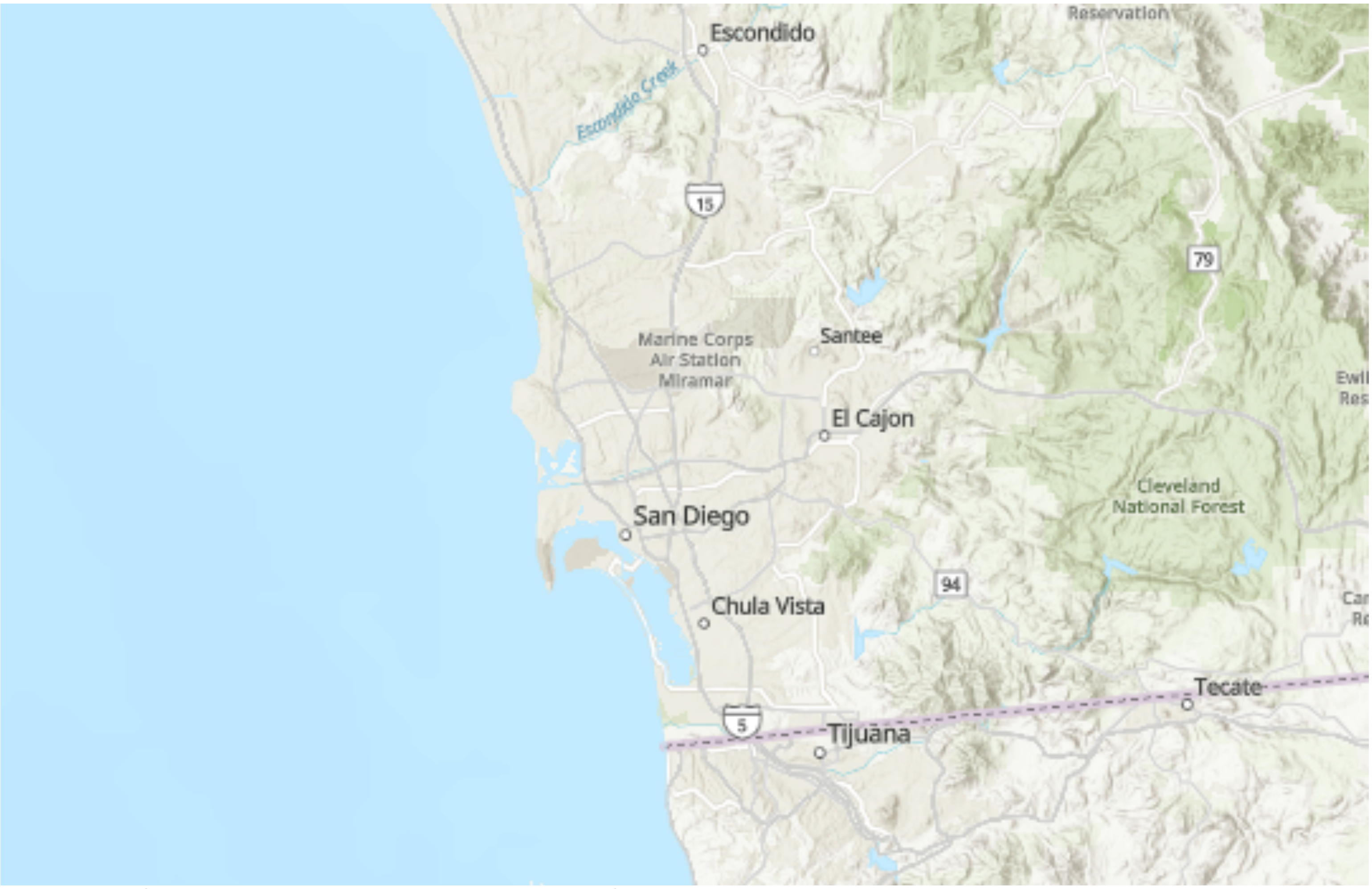
We found that **Otay Mesa's** industrial businesses are located a median distance of **1.06 miles** from the Ports of Entry, compared to **20.07 miles** for the rest of the county.

Otay Mesa does not have a unique level of **Freight Corridor Overlap**; **Kearny Mesa** businesses have more overlapping areas.

Across San Diego, industrially zoned land generally overlaps strongly with freight corridors, so **Otay Mesa** is **not uniquely freight-coupled**.

Although **Otay Mesa has nearly 100%** of its industrial land within the 1**-mile freight corridor**, **Kearny Mesa is almost the same**, and Otay Mesa has the **lowest** per-tract industrial-zoning and coupling-core shares.

Otay Mesa's clearest distinctions are its **proximity to the ports of entry** and a more **logistics-specialized business mix –** among the four areas it has the highest share of border-logistics businesses (about **28%**). The coupling-core measure here is a **three-layer overlap** (industrial zoning combined with General Plan land use, within the 1-mile freight corridor); only the **SANDAG employment-center layer** did not load.



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Limitations

Industrial Polygon Identification

1. Industrial polygons are based on geocoded business license records (manufacturing, wholesale, transportation & warehousing NAICS codes) rather than official zoning, as San Diego's industrial zones aren't strategically located near the US–Mexico ports of entry.
2. Business license addresses may reflect registered offices, not actual warehouse/logistics operations.
3. Ports of entry are only identified at the border and do not

include other potential ports of entry, such as seaports, airports, etc.

4. Border data on one side only. The analysis parameter only covers the US side. Tijuana's maquiladoras and PYME zones, the other half of the cross-border system, are excluded.

The ZAPP layer is viewable at San Diego Zoning and Parcel Information Portal.

Conclusion

Otay Mesa is distinctive because it is close to the border and ports of entry, but the broader pattern of industrial activity near freight corridors is not unique to Otay Mesa. The evidence suggests that freight access is a general feature of San Diego's industrial geography, while Otay Mesa's main difference is its direct border position.